

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
KD2_SAURABH_98912>select o.onum,o.amt,o.odate,o.cnum,o.snum from orders o join customers c on o.snum=c.snum where o.onum in (select onum from orders where c.cname='cisneros');
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3006	1098.16	1990-10-03	2008	1007

2 rows in set (0.00 sec)

7. Extract all the orders of Motika.

```
KD2_SAURABH_98912>SELECT * FROM orders WHERE snum = (SELECT snum FROM salespeople WHERE sname = 'Motika');
```

onum	amt	odate	cnum	snum
3002	1900.10	1990-10-03	2007	1004

1 row in set (0.00 sec)

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8. Find all the orders of salespeople servicing customers in London.

```
KD2_SAURABH_98912>select o.onum,o.amt,o.odate,o.cnum,o.snum from orders o join salespeople s on o.snum=s.snum where city='london';
```

onum	amt	odate	cnum	snum
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3008	4723.00	1990-10-04	2006	1001
3011	9891.88	1990-10-04	2006	1001

4 rows in set (0.00 sec)

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
KD2_SAURABH_98912>SELECT * FROM orders WHERE amt > ANY (SELECT amt FROM orders WHERE cnum IN (SELECT cnum FROM customers WHERE city = 'London'));
```

onum	amt	odate	cnum	snum
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3011	9891.88	1990-10-04	2006	1001

6 rows in set (0.00 sec)

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of Serres salesperson).

```
KD2_SAURABH_98912>select * from customers where rating >= any(select min(rating) from customers where snum=1002);
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.00 sec)

11. Select customers who have a greater rating than any other customer in Rome.

```
KD2_SAURABH_98912>select * from customers where rating > any(select min(rating) from customers where city='rome');
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.00 sec)

12. Select all orders that had amounts that were greater than at least one of the orders from '1990-10-04'.

```
KD2_SAURABH_98912>select * from orders where amt > any(select min(amt) from orders where odate='1990-10-04');
```

onum	amt	odate	cnum	snum
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002
3011	9891.88	1990-10-04	2006	1001

8 rows in set (0.00 sec)

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13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
KD2_SAURABH_98912>select * from orders where amt < (select max(o.amt) from orders o join customers c on o.snum=c.snum where c.city='san jose');
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3007	75.75	1990-10-04	2004	1002
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002

8 rows in set (0.00 sec)

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
KD2_SAURABH_98912>SELECT o.amt, c.cname, c.rating FROM orders o, customers c WHERE o.cnum = c.cnum AND o.amt > (SELECT AVG(amt) FROM orders);
```

amt	cname	rating
5160.45	Liu	200
9891.88	Clemens	100
4723.00	Clemens	100

3 rows in set (0.00 sec)

14. Select those customers whose rating are higher than every customer in London

```
KD2_SAURABH_98912>select * from customers where rating > (select min(rating) from customers where city='london');
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.00 sec)

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
KD2_SAURABH_98912>SELECT * FROM salespeople s WHERE s.snum <> ALL (SELECT c.snum FROM customers c WHERE c.city = s.city);
```

snum	sname	city	comm
1004	Motika	London	0.11
1007	Rifkin	Barcelona	0.15
1003	Axelrod	New York	0.10

3 rows in set (0.00 sec)

9. Find names and numbers of all salesperson who have more than one customer.

```

ERROR 1054 (42S22): Unknown column 'cnum' in 'having clause'
KD2_SAURABH_98912>select sname,snum from salespeople where snum in (select snum from customers group b
y snum having count(cnum)>1);
+-----+-----+
| sname | snum |
+-----+-----+
| Peel  | 1001 |
| Serres| 1002 |
+-----+-----+
2 rows in set (0.01 sec)

```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```

KD2_SAURABH_98912>SELECT snum, sname, city, (SELECT COUNT(*) FROM customers c WHERE c.snum = s.snum) A
S customer_count FROM salespeople s WHERE (SELECT COUNT(*) FROM customers c WHERE c.snum = s.snum) > 1
;
+-----+-----+-----+-----+
| snum | sname | city   | customer_count |
+-----+-----+-----+-----+
| 1001 | Peel  | London |                2 |
| 1002 | Serres| San Jose |                2 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```

KD2_SAURABH_98912>select o.snum,sum(o.amt) as total_amt from orders o group by o.snum having sum(o.am
t)>(select max(amt) from orders);
+-----+-----+
| snum | total_amt |
+-----+-----+
| 1001 | 15382.07 |
+-----+-----+
1 row in set (0.00 sec)

```